

thermal, morphological, structural, Mechanical, optical, electrical and environmental properties of polymers, properties and applications of commodity, engineering and specialty polymers, Natural and synthetic rubbers, Polymer blends and composites, bio polymers, Polymer processing: Compression molding, injection molding, blow molding, extrusion, rotational molding, thermoforming, rubber processing.

2. Department of Applied Physics

Discipline: Physics

Section 1: Mathematical Physics

Vector calculus; matrices; similarity transformations, diagonalization, eigenvalues and eigenvectors; linear differential equations: second order linear differential equations and solutions involving special functions; Partial differential equations, complex analysis: Cauchy-Riemann conditions, Cauchy's theorem, singularities, residue theorem and applications; Laplace transform, Fourier series & analysis; Tensors; Numerical methods.

Section 2: Classical Mechanics

Lagrangian formulation: D'Alembert's principle, Euler-Lagrange equation, Hamilton's principle, symmetry and conservation laws; central force motion: Kepler problem and Rutherford scattering; Periodic motion: small oscillations; coupled oscillations and normal modes; rigid body dynamics; Liouville's theorem; canonical transformations, Poisson brackets, Hamilton-Jacobi equation. Special theory of relativity: Lorentz transformations, relativistic kinematics, mass-energy equivalence.

Section 3: Electromagnetism

Electrostatics and magneto statics, boundary value problems, multipole expansion, Fields in conducting, dielectric, diamagnetic and paramagnetic media, Faraday's law and time varying fields; displacement current; Maxwell's equations; energy and momentum of electromagnetic fields; Propagation of plane electromagnetic waves, reflection, refraction; Electromagnetic waves in dispersive and conducting media.

Section 4: Quantum Mechanics

Wave-particle duality, uncertainty principle; Schrodinger equation; linear vectors and operators in Hilbert space; one dimensional potentials: step potential, finite rectangular well, tunnelling from a potential barrier, particle in a box, harmonic oscillator; two and three dimensional systems: concept of degeneracy; hydrogen atom; Stern-Gerlach experiment, angular momentum and spin; addition of angular momenta; variational method and WKB approximation, time independent perturbation theory; elementary scattering theory, Born approximation; symmetries in quantum mechanical systems; Identical particles; Pauli exclusion principle.

Section 5: Thermodynamics and Statistical Physics

Laws of thermodynamics and their consequences; Macro states and microstates; phase space; ensembles; partition function, free energy, calculation of thermodynamic quantities; classical and quantum statistics; degenerate Fermi gas; black body radiation and Planck's distribution law; Bose-Einstein condensation; first and second order phase transitions, phase equilibria, critical point.

Section 6: Atomic and Molecular Physics

Quantum states of an electron in an atom; Spectra of one-and many-electron atoms; spin-orbit interaction: LS and jj couplings; fine and hyperfine structures; Zeeman and Stark effects; Electronic, rotational and vibrational spectra of diatomic molecules; selection rules; electronic transitions in diatomic molecules, Franck-Condon principle; Raman effect; EPR, NMR, ESR, X-ray spectra; lasers: Einstein coefficients, population inversion, two, three and four level systems.

Section 7: Solid State Physics

Bravais lattices; crystal structures, Miller indices, diffraction methods for structure determination; Reciprocal lattice, bonding in solids; Defects and dislocations; lattice vibrations and thermal properties of solids; free electron theory; band theory of solids; Optical properties of solids; dielectric properties of solid; dielectric function, polarizability, ferroelectricity; magnetic

properties of solids; domains and magnetic anisotropy; superconductivity: Type-I and Type-II superconductors, Meissner effect, London equation, BCS Theory, flux quantization, Josephson junctions.

Section 8: Electronics

Semiconductors, electron and hole statistics in intrinsic and extrinsic semiconductors; Hall effect, metal-semiconductor junctions; Ohmic and rectifying contacts; PN diodes, bipolar junction transistors, field effect devices; negative and positive feedback circuits; oscillators, operational amplifiers, Opto-electronic devices, Microprocessor and micro controller basics.

Section 9: Nuclear and Particle Physics

Basic nuclear properties; Nuclear radii and charge distributions, nuclear binding energy, electric and magnetic moments; semi-empirical mass formula; nuclear models; liquid drop model, nuclear shell model; nuclear force and two nucleon problem; alpha decay, beta-decay, electromagnetic transitions in nuclei; Rutherford scattering, nuclear reactions, conservation laws; fission and fusion; particle accelerators and detectors; elementary particles, Quark model.

Section 10: Optics and Fiber Optics

Interference, diffraction, polarization, Guided wave Optics, Guided Wave Structures, Ray analysis, Modes of planar waveguide, Mode theory for optical fibers, Propagation characteristics of step index fibers, graded index fibers, Signal degradation in optical fiber due to dispersion and attenuation, Optical Sources and detectors for optical fiber communication.

Discipline: Engineering Physics

Section 1: Engineering Mathematics

Linear Algebra: Vector space, basis, linear dependence and independence, matrix algebra, eigenvalues and eigenvectors, rank, solution of linear equations- existence and uniqueness.

Calculus: Mean value theorems, theorems of integral calculus, evaluation of definite and improper integrals, partial derivatives, maxima and minima, multiple integrals, line, surface and volume integrals, Taylor series.

Differential Equations: First order equations (linear and nonlinear), higher order linear differential equations, Cauchy's and Euler's equations, methods of solutions in variation of parameters, complementary function and particular integral, partial differential equations, variable separable method, initial and boundary value problems.

Vector Analysis: Vectors in plane and space, vector operations, gradient, divergence and curl, Gauss's, Green's and Stokes' theorems.

Complex Variables: Analytic functions, Cauchy's integral theorem, Cauchy's integral formula, sequences, series, convergence tests, Taylor and Laurent series, residue theorem.

Section 2: Computational Physics

Roots of Non-linear Equation: Roots of equations, Bisection method, Regula Falsi Method or Method of False position, Secant, Newton-Raphson method, Convergence of these methods.

Interpolation: Finite differences and difference operators, Interpolation with equally spaced data points: Newton's forward and backward formulae for interpolation, Central difference: Gauss forward, Gauss Backward, Stirling, Bessels, Everett's formula for interpolation, Interpolation with unequally spaced data points: Divided differences and their property, Newton Divided differences formula.

Integration: Newton-cotes integration formulae, trapezoidal method, Simpson's 1/3-rule, Simpson's 3/8-rule, Boole's and Weddle's Rule.

Numerical solution of ordinary differential equations: Taylor's series method, Picard's method of successive approximation methods, Euler's method, modified Euler's method, Runge-Kutta method, solution of second order and simultaneous differential equations, Application of optimization and variational methods to problem of interest in applied physics.

Section 3: Quantum Mechanics

Wave-particle duality, uncertainty principle; Schrodinger equation; linear vectors and operators in Hilbert space; one dimensional